

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular & Supplementary Summer 2024**

**Course: B. Tech.**

**Semester: VIII**

**Branch: Mechanical Engineering / Mechanical Engineering (Sandwich) / Production  
Engineering**

**Subject Code & Name: BTMEC801A Fundamentals of Automotive Systems**

**Max Marks: 60**

**Date: 13/06/2024**

**Duration: 3 Hr.**

**Instructions to the Students:**

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

|                                                                                                                                                                                                             | (Level/CO)    | Marks     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                 |               | <b>12</b> |
| A) Explain the phenomena of knock in SI engines.                                                                                                                                                            | CO1           | 6         |
| B) What is an engine block? Explain wet sleeve and dry sleeve inserts with neat sketches.                                                                                                                   | CO1, CO2, CO3 | 6         |
| C) Define the following with respect to engine performance.<br>a) Indicated power                      b) Indicated thermal efficiency<br>c) Brake thermal efficiency              d) Mechanical efficiency | CO1, CO2, CO3 | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                  |               | <b>12</b> |
| A) Discuss the various functions of a camshaft in an IC engine.                                                                                                                                             | CO1, CO2, CO3 | 6         |
| B) With neat sketches explain different phases of Combustion in CI Engine.                                                                                                                                  | CO2           | 6         |
| C) Explain in detail the emission control systems used in IC engine.                                                                                                                                        | CO1           | 6         |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                                                                                 |               | <b>12</b> |
| A) What is a Clutch? Explain Construction and working of Single Plate clutch with sketch.                                                                                                                   | CO2, CO3, CO4 | 6         |
| B) Explain the working of a synchromesh gear box used in Automobiles, with neat sketch.                                                                                                                     | CO2, CO3, CO4 | 6         |
| C) Explain the requirements for an efficient automobile clutch.                                                                                                                                             | CO2, CO3, CO4 | 6         |
| <b>Q.4 Solve Any Two of the following.</b>                                                                                                                                                                  |               | <b>12</b> |
| A) Explain with neat sketch working of Macpherson Strut Suspension system.                                                                                                                                  | CO2, CO3,     | 6         |
| B) Explain with neat sketch the components and working of a Drum Brake.                                                                                                                                     | CO2, CO3,     | 6         |
| C) What is ABS? Discuss with a neat sketch the operation Methodology of ABS.                                                                                                                                | CO2, CO3,     | 6         |
| <b>Q. 5 Solve Any Two of the following.</b>                                                                                                                                                                 |               | <b>12</b> |
| A) Explain with neat sketch working of hydraulic power steering system.                                                                                                                                     | CO2, CO3, CO4 | 6         |
| B) Explain the terms (i) Camber angle (ii) Castor Angle (iii) Steering axis Inclination (iv) Toe-In Toe-Out.                                                                                                | CO2, CO3, CO4 | 6         |
| C) Differentiate between Tyres with tube and Tubeless tyres.                                                                                                                                                | CO2, CO3      | 6         |

\*\*\* End \*\*\*